Infinite geometric series

1 What condition must be satisfied by an infinite geometric series in order for its sum to exist?

2 For each of the following infinite geometric sequences:

i Find the common ratio.

ii Determine whether the sum is divergent or convergent.

a $11, 22, 44, 88, \ldots$

b $3, -12, 48, -192, \ldots$

c $-40, -20, -10, -5, \ldots$

d $256, 64, 16, 4, \ldots$

3 For each of the following infinite geometric sequences:

i Find the common ratio, r.

ii Find the limiting sum of the series.

a $2, \frac{1}{2}, \frac{1}{8}, \frac{1}{32}, \ldots$

b $125, 25, 5, 1, \ldots$

c $16, -8, 4, -2, \ldots$

d $28, -7, \frac{7}{4}, -\frac{7}{16}, \ldots$

4 Consider the infinite geometric series: $5 + \sqrt{5} + 1 + \ldots$

a Find the common ratio, r, expressing your answer with a rational denominator.

b Find the limiting sum, expressing your answer with a rational denominator.

5 Consider the infinite geometric series: $6 + 2 + \frac{2}{3} + \frac{2}{9} + \ldots$

a Find the number of terms, n, that would be required to give a sum of $\frac{177\,146}{19\,683}$.

b Find the infinite sum of the series.

6 Find the limiting sum of the infinite series $\frac{1}{5} + \frac{3}{5^2} + \frac{1}{5^3} + \frac{3}{5^4} + \frac{1}{5^5} + \ldots$

7 For a particular geometric sequence, $u_1 = 2$ and $S_\infty = 4$.

a Find r, the common ratio.

b Find the first 3 terms in the sequence.

8 Consider the recurring decimal $0.2222\ldots$ in the form

$$\frac{2}{10} + \frac{2}{100} + \frac{2}{1000} + \frac{2}{10\,000} + \ldots$$

Rewrite the recurring decimal as a fraction.

9 Find the value of the following:

a $\displaystyle\sum_{i=1}^{\infty} 3\left(\frac{1}{4}\right)^{i-1}$

b $\displaystyle\sum_{i=1}^{\infty} 5\left(-\frac{1}{4}\right)^{i-1}$

c $\displaystyle\sum_{i=1}^{\infty} \frac{1}{8}\left(-\frac{1}{2}\right)^{i-1}$

d $\displaystyle\sum_{i=1}^{\infty} -\frac{1}{2}\left(\frac{5}{7}\right)^{i-1}$

e $\displaystyle\sum_{i=1}^{\infty} (0.9)^{i}$

f $\displaystyle\sum_{k=1}^{\infty} 9^{-k}$

Recurring decimals

10 For each of the following recurring decimals:

i Rewrite the decimal as a geometric series.

ii Express the decimal as a fraction.

a $0.6666\ldots$

b $0.066\,66\ldots$

c $0.151\,515\,15\ldots$

d $0.021\,212\,1\ldots$

11 Consider the recurring decimal $0.575\,75\ldots$ in the form

$$0.57 + 0.0057 + \ldots$$

Rewrite the recurring decimal as a fraction.

12 Consider the sum $0.25 + 0.0025 + 0.000\,025 + \ldots$

a Express the sum as a decimal.

b Hence, express the decimal as a fraction.

13 Find the value of $0.994 + 0.000\,994 + 0.000\,000\,994\ldots$

Algebraic series

14 Consider a geometric series in which all terms are positive that has first term a and common ratio r. The sum of the infinite series is 768. The sum of the first 4 terms is 765.
Find the first three terms of the sequence.

15 The limiting sum of the infinite sequence $1,\ 4x,\ 16x^2, \ldots$ is 5. Find the value of x.

16 Consider the infinite sequence: $1,\ x-7,\ (x-7)^2,\ (x-7)^3$